

AI- ASSISTANT CODING
ASSIGNMENT 5.1

2303A51653

V. Sai Akshaya

BATCH 28

Task Description #1 – Privacy in API Usage

TASK

Use an AI tool to generate a Python program that connects to a weather API securely without exposing API keys.

PROMPT

Generate code to fetch weather data securely without exposing API keys in the code.

CODE

 Insecure Version (Hardcoded API Key)

```
import requests
```

```
API_KEY = "12345ABCDE" # Hardcoded API key (NOT SAFE)
```

```
CITY = "Hyderabad"
```

```
url = f"https://api.openweathermap.org/data/2.5/weather?q={CITY}&appid={API_KEY}"
```

```
response = requests.get(url)
```

```
print(response.json())
```

 Secure Version (Using Environment Variables)

```
import requests
```

```
import os
```

```
API_KEY = os.getenv("WEATHER_API_KEY") # Read from environment variable
```

```
CITY = "Hyderabad"
```

```
url = f"https://api.openweathermap.org/data/2.5/weather?q={CITY}&appid={API_KEY}"
```

```
response = requests.get(url)
```

```
print(response.json())
```

OUTPUT

- Weather details of the given city are displayed in JSON format.
- API key is **not visible** in the code.

EXPLANATION

- Hardcoding API keys is a **privacy risk**.
 - Environment variables keep keys hidden and secure.
 - This approach prevents accidental exposure on GitHub or public files.
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Task Description #2 – Privacy & Security in File Handling

TASK

Store user data securely by avoiding plain-text password storage.

PROMPT

Generate a Python script that stores user data securely instead of plain text.

CODE

Insecure Version (Plain Text Password)

```
name = input("Enter name: ")
email = input("Enter email: ")
password = input("Enter password: ")

with open("users.txt", "a") as file:
    file.write(f"{name},{email},{password}\n")
```

Secure Version (Hashed Password)

```
import hashlib

name = input("Enter name: ")
email = input("Enter email: ")
password = input("Enter password: ")

hashed_password = hashlib.sha256(password.encode()).hexdigest()
```

with open("users.txt", "a") as file:

```
file.write(f"{name},{email},{hashed_password}\n")
```

OUTPUT

- User data is stored.
- Password appears as a **hashed value**, not readable text.

EXPLANATION

- Plain text passwords can be stolen easily.
 - Hashing converts passwords into irreversible values.
 - Even if the file is leaked, passwords remain protected.
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Task Description #3 – Transparency in Algorithm Design

TASK

Create an Armstrong number checker with transparent explanation.

PROMPT

Generate Python code to check an Armstrong number and explain it line by line.

CODE

```
def is_armstrong(number):  
    digits = list(map(int, str(number)))  
    power = len(digits)  
  
    total = sum(d ** power for d in digits)  
  
    return total == number  
  
num = int(input("Enter a number: "))  
print(is_armstrong(num))
```

OUTPUT

Enter a number: 153

True

EXPLANATION

- Number is converted into digits.

- Each digit is raised to the power of total digits.
 - If the sum equals the original number, it is an Armstrong number.
 - Code and explanation match clearly, ensuring transparency.
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Task Description #4 – Transparency in Algorithm Comparison

TASK

Implement and compare QuickSort and BubbleSort.

PROMPT

Generate Python code for QuickSort and BubbleSort with step-by-step explanations.

CODE

Bubble Sort

```
def bubble_sort(arr):  
    for i in range(len(arr)):  
        for j in range(0, len(arr)-i-1):  
            if arr[j] > arr[j+1]:  
                arr[j], arr[j+1] = arr[j+1], arr[j]  
    return arr
```

Quick Sort

```
def quick_sort(arr):  
    if len(arr) <= 1:  
        return arr  
  
    pivot = arr[0]  
    left = [x for x in arr[1:] if x <= pivot]  
    right = [x for x in arr[1:] if x > pivot]  
  
    return quick_sort(left) + [pivot] + quick_sort(right)
```

OUTPUT

Bubble Sort: [1, 2, 4, 5, 8]

Quick Sort: [1, 2, 4, 5, 8]

EXPLANATION

- BubbleSort compares adjacent elements repeatedly.
- QuickSort divides the list using a pivot.
- BubbleSort is slower ($O(n^2)$).
- QuickSort is faster ($O(n \log n)$).

Aspect	Bubble Sort	Quick Sort
Method	Swapping adjacent elements	Divide and conquer
Time Complexity	$O(n^2)$	$O(n \log n)$
Efficiency	Slow	Fast
Use Case	Small lists	Large datasets

Task Description #5 – Transparency in AI Recommendations

TASK

Create a recommendation system with explainable suggestions.

PROMPT

Generate a recommendation system that explains why each product is suggested.

CODE

```
def recommend_products(user_interest):  
    products = {  
        "fitness": ["Dumbbells", "Yoga Mat"],  
        "technology": ["Laptop", "Smartphone"],  
        "books": ["Fiction Novel", "Self-help Book"]  
    }  
  
    recommendations = products.get(user_interest, [])  
  
    for item in recommendations:  
        print(f'Recommended: {item} because it matches your interest in {user_interest}.')
```

OUTPUT

Recommended: Laptop because it matches your interest in technology.

Recommended: Smartphone because it matches your interest in technology.

EXPLANATION

- Recommendations are based on user interests.
 - Each suggestion includes a clear reason.
 - Improves transparency and user trust.
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